

Zepto E-Commerce Inventory SQL Analysis

1. Project Background and Objective

This project focuses on analysing Zepto's e-commerce inventory data using SQL to understand product performance, inventory availability, pricing strategies, and category-level business trends.

The main objective of this project is to explore inventory data and identify important insights related to product pricing, discounts, stock availability, revenue potential, and product value.

By analysing inventory information, this project aims to help the business improve inventory management, optimize pricing decisions, identify revenue opportunities, and maintain better product availability for customers.

2. Understanding the Dataset

The dataset represents Zepto's product inventory information collected from an e-commerce platform. Each record represents a unique Stock Keeping Unit (SKU) containing product details, pricing information, inventory levels, and availability status.

The dataset contains information related to:

- Product SKU identification
- Product categories
- Product names
- Maximum Retail Price (MRP)
- Discount percentage
- Discounted selling price
- Available inventory quantity
- Product weight
- Stock availability status
- Product quantity details

The SQL database table was created with important attributes including SKU ID, category, product name, pricing fields, quantity information, and stock availability indicators.

This information helps analyse product performance and understand inventory-related business problems.

3. Data Exploration and Data Cleaning Process

Before performing business analysis, the raw inventory data was explored and cleaned using SQL.

The data preparation process included:

- Checking the total number of records available in the database
- Reviewing sample records to understand dataset structure
- Identifying missing or null values
- Analysing available product categories
- Comparing in-stock and out-of-stock products
- Identifying duplicate product names with multiple SKUs

The SQL exploration process helped understand the quality and structure of the inventory dataset before analysis.

Data cleaning activities included:

- Removing invalid records where product prices were zero
- Converting pricing values from paise into rupees for accurate business interpretation

These steps ensured that the dataset was reliable and ready for further analysis.

4. Inventory Analysis Using SQL

After preparing the dataset, SQL queries were developed to answer important business questions and identify inventory-related insights.

The analysis focused on understanding:

- Products offering the highest discounts
- High-value products that are unavailable
- Revenue potential across product categories
- Products with high prices and low discounts
- Categories with better discount strategies
- Product value based on price per gram
- Inventory weight distribution across categories

These analyses help the business understand product performance and improve inventory planning.

5. Business Questions and SQL Analysis Results

5.1 Top 10 Best Value Products Based on Discount Percentage

Business Question:

Which products provide the highest discounts compared with their original MRP?

Purpose:

This analysis helps identify products with attractive discount offers that may influence customer purchase decisions.

SQL Analysis:

The query ranks products based on discount percentage and identifies the top discounted products.

Result Screenshot:

	name character varying (150) 🔒	mrp numeric (8,2) 🔒	discountpercent numeric (5,2) 🔒
1	Dukes Waffy Orange Wafers	4500.00	51.00
2	Dukes Waffy Strawberry Wafers	4500.00	51.00
3	Dukes Waffy Chocolate Wafers	4500.00	51.00
4	Epigamia Fruit Yogurt Alphonso Mango	4000.00	50.00
5	Epigamia Fruit Yogurt Vanilla	4000.00	50.00
6	Chef's Basket Durum Wheat Elbow Pasta	16000.00	50.00
7	Chef's Basket Durum Wheat Penne Pasta	16000.00	50.00
8	Ceres Foods Nalli Nihari Instant Liquid Masala	22000.00	50.00
9	Chef's Basket Durum Wheat Fusilli Pasta	16000.00	50.00
10	Moi Soi Sichuan Chilli Oil- For Stir Fry Marinade Spread & ...	28000.00	50.00

5.2 Products with High MRP but Currently Out of Stock

Business Question:

Which expensive products are currently unavailable?

Purpose:

Identifying high-value out-of-stock products helps the company prioritize inventory replenishment and avoid losing potential revenue.

SQL Analysis:

Products with MRP greater than ₹300 and unavailable stock were identified.

Result Screenshot:

	category character varying (120) 🔒	total_revenue numeric 🔒
1	Fruits & Vegetables	1084600.00
2	Meats, Fish & Eggs	2069300.00
3	Biscuits	2500760.00
4	Beverages	5505100.00
5	Dairy, Bread & Batter	5505100.00
6	Health & Hygiene	6418000.00
7	Home & Cleaning	12266100.00
8	Ice Cream & Desserts	22438500.00
9	Chocolates & Candies	22438500.00
10	Packaged Food	22438500.00
11	Personal Care	27084900.00
12	Paan Corner	27084900.00
13	Munchies	33736900.00
Total rows: 14		Query complete 00:00:00.110

5.4 Expensive Products with Low Discounts

Business Question:

Which products have high prices but provide limited discounts?

Purpose:

This helps evaluate pricing strategies and identify products where discount adjustments may improve competitiveness.

SQL Analysis:

Products with MRP above ₹500 and discount percentage below 10% were identified.

Result Screenshot:

	name character varying (150)	mrp numeric (8,2)	discountpercent numeric (5,2)
1	Dhara Kachi Ghani Mustard Oil Jar	125000.00	8.00
2	Saffola Gold (Jar)	124000.00	0.00
3	Fortune Rice Bran Health Oil (Jar)	105000.00	1.00
4	Dhara Filtered Groundnut Oil (Jar)	105000.00	1.00
5	Dhara Filtered Groundnut Oil (Jar)	105000.00	0.00
6	Fortune Soyabean Oil	100500.00	0.00
7	Fortune Sunlite Refined Sunflower (Jar)	92500.00	0.00
8	Surf Excel Matic Powder Front Load	81000.00	7.00
9	Surf Excel Matic Top Load	72000.00	9.00
10	Pedigree Puppy Dry Dog Food Food Chicken & Milk	69000.00	6.00
11	Pedigree Dog Food Adult Meat & Rice	66000.00	7.00
12	Lizol Double Concentrate Disinfectant Floor Cleaner Citrus	65000.00	8.00
13	Nestle Nestle Nan Pro Follow-Up Formula	65000.00	0.00
Total rows: 1142		Query complete 00:00:00.107	

5.5 Categories Offering Highest Average Discounts

Business Question:

Which product categories provide the highest average discounts?

Purpose:

This analysis helps evaluate category-level discount strategies.

SQL Analysis:

Average discount percentage was calculated for each category and ranked accordingly.

Result Screenshot:

	category character varying (120) 🔒	avg_discount numeric 🔒
1	Fruits & Vegetables	15.46
2	Meats, Fish & Eggs	11.03
3	Ice Cream & Desserts	8.32
4	Chocolates & Candies	8.32
5	Packaged Food	8.32

5.6 Product Value Analysis Using Price Per Gram

Business Question:

Which products provide better value based on price compared with their weight?

Purpose:

This analysis identifies cost-effective products by calculating price per gram.

SQL Analysis:

Products above 100 grams were analysed based on discounted selling price per gram.

Result Screenshot:

	name character varying (150)	weightingms integer	discountedsellingprice numeric (8,2)	price_per_gram numeric
1	Vicks Cough Drops Menthol	1160	2000.00	1.72
2	Aashirvaad Iodised Salt	1000	1900.00	1.90
3	Onion	3000	5700.00	1.90
4	Onion	1000	2100.00	2.10
5	Tata Salt	1000	2400.00	2.40
6	Shubh kart - Nirmal sugandhi mogra wet dhoop zipper 20 sticks	1160	2800.00	2.41
7	Beetroot	500	1300.00	2.60
8	Shubh kart - Tejas Twisted Cotton Wicks 1000n	1000	2800.00	2.80
9	Potato	3000	8400.00	2.80
10	Potato	1000	2900.00	2.90
11	Carrot	500	1500.00	3.00
12	Baby Potato	500	1600.00	3.20
13	Raw Banana	500	1700.00	3.40
Total rows: 1395		Query complete 00:00:00.119		

5.7 Product Weight Category Classification

Business Question:

How can products be grouped according to their weight?

Purpose:

Weight segmentation helps understand product packaging and inventory distribution.

Products were classified into:

- Low Weight
- Medium Weight
- Bulk Weight

using SQL CASE statements.

Result Screenshot:

	name character varying (150)	weightingms integer	weight_category text
1	Popular Essentials Poppy Seeds (Gasagase)	100	Low
2	Origami Kitchen Towels 2 In 1 Quilted 2ply	116	Low
3	Origami Good Karma Paper Serviettes - 100 napkins (1 ply)	58	Low
4	Johnson's Baby Top To Toe Bath Body Wash	210	Low
5	Harpic Hygienic Toilet Cleaner Rim Block Citrus	26	Low
6	Lizol Disinfectant Surface & Floor Cleaner Liquid Jasmine	500	Low
7	Sapota	348	Low
8	Britannia Treart Creme Wafers Chocolate	60	Low
9	Popular Essentials Dhania (Coriander Seeds)	200	Low
10	Popular Essentials Californian Almond	250	Low
11	Everest Shahi Biryani Masala	50	Low
12	Keya Italian Pizza Oregano	80	Low
13	Figaro Plain Green Olive	450	Low
Total rows: 1783		Query complete 00:00:00.124	

5.8 Total Inventory Weight by Category

Business Question:

Which categories contain the highest total inventory weight?

Purpose:

This analysis helps understand storage requirements and category-level inventory distribution.

SQL Analysis:

Total inventory weight was calculated using product weight multiplied by available quantity.

Result Screenshot:

	category character varying (120) 🔒	total_weight bigint 🔒
1	Meats, Fish & Eggs	48016
2	Biscuits	84431
3	Fruits & Vegetables	91794
4	Health & Hygiene	142904
5	Dairy, Bread & Batter	143735
6	Beverages	143735
7	Paan Corner	348187
8	Personal Care	348187
9	Home & Cleaning	373236
10	Ice Cream & Desserts	490797
11	Chocolates & Candies	490797
12	Packaged Food	490797
13	Munchies	1404654
14	Cooking Essentials	1404654

6. Business Insights and Recommendations

Based on the SQL analysis results, the company can focus on the following improvements:

- Maintain sufficient stock levels for high-value products to reduce missed sales opportunities.
- Review discount strategies for different product categories to improve customer attraction.
- Monitor high-performing categories to optimize inventory allocation.
- Use price-per-gram analysis to promote value-for-money products.
- Improve inventory planning by analysing product weight and quantity distribution.
- Regularly monitor out-of-stock products to maintain better customer experience.

These recommendations can help improve inventory efficiency, pricing decisions, and overall e-commerce performance.

7. Project Documentation and SQL Repository

All project components, including:

- SQL database creation scripts
- Data exploration queries
- Data cleaning queries
- Business analysis queries
- Query output screenshots

will be organised in a structured project repository.

Proper documentation ensures that the project can be easily reviewed, maintained, and presented as a complete SQL Data Analyst portfolio project.